

NOTES: CLASS-3

1. Find the Fourier Series of $f(x) = x + x^2$ in $(-\pi, \pi)$.

Hence deduce that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

Solution : The Fourier expansion of $f(x)$ over $(-\pi, \pi)$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \dots \dots \dots (1)$$

where

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx.$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x + x^2 dx$$

$$= \frac{1}{\pi} \left[\frac{x^2}{2} + \frac{x^3}{3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{2} + \frac{\pi^3}{3} - \frac{\pi^2}{2} + \frac{\pi^3}{3} \right]$$

$$\Rightarrow a_0 = \frac{2\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \cos nx dx$$

$$= \frac{1}{\pi} \left[(x + x^2) \left(\frac{\sin nx}{n} \right) - (1 + 2x) \left(-\frac{\cos nx}{n^2} \right) + 2 \left(-\frac{\sin nx}{n^3} \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[(1 + 2x) \left(\frac{\cos nx}{n^2} \right) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi n^2} [(1 + 2\pi) \cos n\pi - (1 + 2(-\pi) \cos n(-\pi))]$$

$$= \frac{1}{\pi n^2} (4\pi) \cos n\pi$$

$$\Rightarrow a_n = \frac{4}{n^2} (-1)^n$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} (x + x^2) \sin nx dx$$

$$= \frac{1}{\pi} \left[(x + x^2) \left(-\frac{\cos nx}{n} \right) - (1 + 2x) \left(-\frac{\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right]_{-\pi}^{\pi}$$

$$\begin{aligned}
&= \frac{1}{\pi} \left[(x + x^2) \left(-\frac{\cos nx}{n} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right]_{-\pi}^{\pi} \\
&= \frac{1}{\pi n^2} \left[(\pi + \pi^2) \left(-\frac{\cos n\pi}{n} \right) + 2 \left(\frac{\cos n\pi}{n^3} \right) - (-\pi + (-\pi)^2) \left(-\frac{\cos n(-\pi)}{n} \right) + 2 \left(\frac{\cos n(-\pi)}{n^3} \right) \right] \\
&= \frac{-1}{\pi n} (2\pi) \cos n\pi \\
&= \frac{-2}{n} (-1)^n
\end{aligned}$$

$$\Rightarrow b_n = \frac{2}{n} (-1)^{n+1}$$

The required Fourier series is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$\Rightarrow x + x^2 = \frac{2\pi^2}{3} \frac{1}{2} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \cos nx + \frac{2}{n} (-1)^{n+1} \sin nx \right)$$

$$\Rightarrow x + x^2 = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left(\frac{4}{n^2} (-1)^n \cos nx + \frac{2}{n} (-1)^{n+1} \sin nx \right) \dots \dots \dots (1)$$

Deduction :

To deduce the series we shall put $x = -\pi$ in the above equation.

Here $f(x) = x + x^2$ is valid for all values of x between $-\pi$ and π but not at the end points since the given interval is open. Therefore, $x = -\pi$ is a point of discontinuity and hence,

$$\begin{aligned}
f(-\pi) &= \frac{1}{2} [f(-\pi - 0) + f(-\pi + 0)] \\
&= \frac{1}{2} [f(\pi - 0) + f(-\pi + 0)] \quad (\text{since } f(x) \text{ is periodic with } 2\pi) \\
&= \frac{1}{2} [\pi + \pi^2 + (-\pi) + (-\pi)^2]
\end{aligned}$$

$$\Rightarrow f(-\pi) = \pi^2 \dots \dots \dots (2)$$

Substituting the value of $f(-\pi)$ from (2) and putting $x = \pi$ in (1) we obtain,

$$\pi^2 = \frac{\pi^2}{3} + 4 \left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots \right]$$

Simplifying, we get,

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots \dots \dots$$

REFERENCES:

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2. Higher Engineering Mathematics-B.V. Ramana
